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F															
UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-22 20kV Central WTP Area SCHEMATIC DIAGRAM SBC-W INCOMING H01															
				Date Prep. 15.11.2018		PROJECT TITLE				ABB EPMV		COVER SHEET		Scale NTS = H01	
				Prepared YR		MV Switchgear for MNA Serang Plant				PT. ABB Sakti Industri		MVD-22 20kV Central WTP Area		Customer PT Multimas Nabati Asahan & 304	
00		Issued for Approval		15.01.19 YR		Checked AW		SCHEMATIC DIAGRAM				Drawing Nbr. 18.042-MNA-304.04		Sh. 1	
Ind.		Revision		Date		Name		Approved AI				Based on: BE318042		Customer Doc. Nbr.	
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					Date Prep.	15.11.2018	PROJECT TITLE			ABB	EPMV	Index of Sheet MVD-22 20kV Central WTP Area SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
					Prepared	YR	MV Switchgear for MNA Serang Plant						Customer	PT Multimas Nabati Asahan	& 304
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	REFERENCE DESIGNATION OF OBJECTS IN ELECTRICAL DOCUMENTS																				-BGB5 POSITION SWITCH OF CIRCUIT-BREAKER SIGNALLING UNDERVOLTAGE RELEASE ENERGIZED																			
B	(IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001)																				-BGB6 POSITION SWITCH OF CIRCUIT-BREAKER SIGNALLING UNDERVOLTAGE RELEASE EXCLUDED MECHANICALLY																			
	DESIGNATION																				DESCRIPTION																			
B	-AA MULTIFUNCTION UNIT (CENTRAL UNIT)																				-BGB7 POSITION SWITCHES OF CIRCUIT-BREAKER ON SWITCHGEAR																			
	-BAD1,-BAD4 CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L1																				-BGB8...-BGB10 POSITION SWITCHES OF CIRCUIT-BREAKER																			
	-BAD2,-BAD5 CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L2																				-BGB11 POSITION SWITCH OF CIRCUIT-BREAKER OPERATING FOR MANUAL OPENING																			
	-BAD3,-BAD6 CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L3																				-BGB12 PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN OPEN POSITION																			
	-BAR VOLTAGE PROTECTION RELAY																				-BGB13 PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN CLOSED POSITION																			
	-BAS1 VOLTAGE SENSOR LOCATED ON PHASE L1																				-BGD POSITION SWITCH OF L.V. INSTRUMENT COMPARTMENT DOOR.																			
	-BAS2 VOLTAGE SENSOR LOCATED ON PHASE L2																				-BGD1 POSITION SWITCH OF CIRCUIT-BREAKER COMPARTMENT DOOR																			
	-BAS3 VOLTAGE SENSOR LOCATED ON PHASE L3																				-BGD2 POSITION SWITCH OF CABLE COMPARTMENT DOOR																			
	-BAT1 VOLTAGE TRANSFORMER LOCATED ON PHASE L1																				-BGE1 POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION																			
	-BAT2 VOLTAGE TRANSFORMER LOCATED ON PHASE L2																				-BGE2 POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN CLOSED POSITION																			
C	-BAT3 VOLTAGE TRANSFORMER LOCATED ON PHASE L3																				-BGE3 POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																			
	-BAT4 VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L1																				-BGE4 POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE ACTUATED BY ELECTRIC MOTOR																			
	-BAT5 VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L2																				-BGE5 POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2 ACTUATED BY ELECTRIC MOTOR																			
	-BAT6 VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L3																				-BGE7 POSITION SWITCH SIGNALLING EARTHING SWITCH NOT IN MANUAL OPERATION																			
	-BCD DIFFERENTIAL PROTECTIVE RELAY																				-BGE8 POSITION SWITCHES OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH- DISCONNECTOR -QBS																			
	-BCF FEEDER PROTECTION RELAY																				-BGE11 POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION																			
	-BCG GENERATOR PROTECTION RELAY																				-BGE12 POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN CLOSED POSITION																			
	-BCM MOTOR PROTECTION RELAY																				-BGE13 POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																			
	-BCN NEUTRAL (RESIDUAL) CURRENT TRANSFORMER																				-BGE14, -BGE15 POSITION SWITCHES FOR OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS, ACTUATED BY ELECTRIC MOTOR																			
	-BCP TRANSFORMER PROTECTION RELAY																				-BGF POSITION SWITCHES OF MEDIUM VOLTAGE FUSES																			
D	-BCR CURRENT PROTECTION RELAY																				-BGI1 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION																			
	-BCS1 CURRENT SENSOR LOCATED ON PHASE L1																				-BGI2 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN CLOSED POSITION																			
	-BCS2 CURRENT SENSOR LOCATED ON PHASE L2																				-BGI3 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																			
	-BCS3 CURRENT SENSOR LOCATED ON PHASE L3																				-BGI4 POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR																			
	-BCT1 CURRENT TRANSFORMER LOCATED ON PHASE L1																				-BGI5 POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR																			
	-BCT2 CURRENT TRANSFORMER LOCATED ON PHASE L2																				-BGI6 POSITION SWITCH SIGNALLING DISCONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN EMERGENCY MANUAL OPERATION																			
	-BCT3 CURRENT TRANSFORMER LOCATED ON PHASE L3																				-BGI7 POSITION SWITCH SIGNALLING DICONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN MANUAL OPERATION																			
	-BCT4,-BCT7 CURRENT TRANSFORMER LOCATED ON PHASE L1																				-BGI8 POSITION SWITCHES OF SWITCH-DISCONNECTOR -QBS																			
	-BCT5,-BCT8 CURRENT TRANSFORMER LOCATED ON PHASE L2																				-BGI9 POSITION SWITCHES OF OPERATION OF SWITCH-DISCONNECTOR -QBS ACTUATED BY ELECTRIC MOTOR																			
	-BCT6,-BCT9 CURRENT TRANSFORMER LOCATED ON PHASE L3																				-BGI11 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION																			
E	-BCZ DISTANCE PROTECTION RELAY																				-BGI12 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN CLOSED POSITION																			
	-BEF FREQUENCY PROTECTION RELAY																				-BGI13 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																			
	-BER SUPERVISION RELAYS																				-BGI14, -BGI15 POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR																			
	-BES SYNCHRONIZING RELAY																				-BGI21 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION																			
	-BET THERMAL PROTECTION RELAY																				-BGI22 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN CLOSED POSITION																			
	-BGA1 POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CIRCUIT-BREAKER COMPARTMENT																				-BGI23 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																			
	-BGA2 POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN BUSBAR COMPARTMENT (SINGLE BUSBAR SYSTEM) OR IN FRONT BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)																				-BGI24, -BGI25 POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR																			
	-BGA3 POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CABLE COMPARTMENT																																							
	-BGA4 POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN REAR BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)																																							
	F	-BGB1...-BGB3 POSITION SWITCHES OF CIRCUIT-BREAKER																																						
-BGB4 POSITION SWITCH OF CIRCUIT-BREAKER: PASSING CONTACT CLOSING MOMENTARILY WHEN THE BREAKER OPENS																																								
				Date Prep.				15.11.2018				PROJECT TITLE				ABB EPMV				REFERENCE DESIGNATIONS				Scale				NTS				= H01								
				Prepared				YR				MV Switchgear for MNA Serang Plant								MVD-22 20kV Central WTP Area				Customer				PT Multimas Nabati Asahan				& 304								
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A	-BGI31	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS							A
	-BGI32	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR							
	-BGI33	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR							
	-BGI41...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.							
B	-BGI42	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT							B
	-BGI43...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION						-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE							
	-BGI51	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION						-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT							
	-BGI52	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB1	HEATER LOCATED IN CABLE COMPARTMENT							
C	-BGI53	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT							C
	-BGK	POSITION SWITCH OF KEY LOCK						-EB3	HEATER LOCATED IN MOTOR CABINET							
	-BGL1	POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB4	HEATER FOR VENTILATION TEST							
	-BGS1...-BGS4	POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.							
D	-BGS5	PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION						-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE							D
	-BGT1	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE							
	-BGT2	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1							
	-BGT3	POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2							
E	-BGT4	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3							E
	-BGT5	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS							
	-BGT6	POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1							
	-BGT7	POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION						-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2							
F	-BGT8	POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3							F
	-BGT11	PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER							
	-BGT12	PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS							
	-BGV1	POSITION SWITCH OF FRONT VENTILATOR						-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1							
G	-BGV2	POSITION SWITCH OF REAR VENTILATOR						-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2							G
	-BM	HYGROSTAT						-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3							
	-BPA4	PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-FCT	THERMAL OVERLOAD RELEASES							
	-BPA5	PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GA	GENERATORS							
H	-BPA6	PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR							H
	-BPA7	PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR							
	-BPS1	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER						-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK							
	-BPS2	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2						-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE							
I	-BPS3	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3						-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE							I
	-BR	FLAME DETECTORS, SMOKE DETECTORS						-KFA3...-KFA9	AUXILIARY RELAYS OR CONTACTORS							
	-BT	THERMOSTAT						-KFC	CLOSING RELAYS OR CONTACTORS							
	-BT1	THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS						-KFI	INTEGRATED CIRCUITS							
J	-BT2	THERMOSTAT FOR ENERGIZATION OF VENTILATORS						-KFL	LOCKOUT RELAY							J
	-BT3	THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM						-KFM	MICROPROCESSORS							
	-BU	BUCHHOLZ RELAY						-KFN	ANTIPUMPING RELAYS							
	-BX1	UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING						-KFO	OPENING RELAYS OR CONTACTORS							
K	-BX2	ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING						-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)							K
	-BZ,-BZ4	MULTIFUNCTION SENSORS						-KFR	RECLOSING RELAYS							
	-BZ1	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1						-KFS	SYNCHRONIZING DEVICES							
	-BZ2	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2						-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS							
L	-BZ3	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3						-KFZ	CONTROLLERS							L
								-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER							
								-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD							
								-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1							
M								-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2							M
								-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS							
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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE	-QBH	MANUAL CIRCUIT-BREAKERS
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2	-QBL	LINKS
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS	-QBM	MINIATURE SWITCH-DISCONNECTORS
B	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING	-QBS	SWITCH-DISCONNECTORS
	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT	-QBT	TRUCK
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER	-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING
C	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS	-QCE	EARTHING SWITCH
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER	-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER	-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)
D	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER	-QQ	CLUTCHES
	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS	-RAA	ANTIFERRORESONANCE RESISTOR
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER	-RAD	DIODES
E	-PFB	BLUE SIGNAL LAMPS	-RAI	INDUCTORS
	-PFF	FLAG RELAYS	-RAR	RESISTORS
	-PFG	GREEN SIGNAL LAMPS	-RF	FILTERS
F	-PFR	RED SIGNAL LAMPS	-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING
	-PFS	SHORT CIRCUIT INDICATORS	-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT
	-PFV	VOLTAGE INDICATORS	-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE
G	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE	-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE	-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD
	-PFW	WHITE SIGNAL LAMPS	-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1
H	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS	-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2
	-PFY	YELLOW SIGNAL LAMPS	-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2
	-PGA	AMMETERS	-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS
I	-PGC	COUNTERS	-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4
	-PGF	FREQUENCYMETERS	-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5
	-PGH	HOURMETERS	-SFA	AMMETRIC SWITCHES
J	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE	-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)	-SFL	LOCKING CONTACTS
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)	-SFO	OPENING PUSH-BUTTONS
K	-PGJ	ACTIVE ENERGY METERS	-SFR	RESET PUSH-BUTTONS
	-PGK	REACTIVE ENERGY METERS	-SFS	SELECTOR SWITCHES
	-PGM	MULTIFUNCTION INDICATORS	-SFT	TEST PUSH-BUTTONS
L	-PGP	POWER-FACTOR METERS	-SFU	UNLOCKING PUSH-BUTTONS
	-PGQ	VARMETERS	-SFV	VOLTMETRIC SWITCHES
	-PGS	SYNCHRONOSCOPES	-TA	POWER TRANSFORMERS
M	-PGV	VOLTMETERS	-TB	CONVERTER
	-PGW	WATTMETERS	-TB1	RECTIFIER FOR FIRST OPENING RELEASE
	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)	-TB2	RECTIFIER FOR SECOND OPENING RELEASE
N	-QAB	CIRCUIT-BREAKERS	-TB3	RECTIFIER FOR CLOSING RELEASE
	-QAC	CONTACTORS (FOR POWER)	-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING
	-QBD	DISCONNECTORS	-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT
O	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)	-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)	-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)		
P	-QBD4	TWO-WAY DISCONNECTOR		
	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER		

				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	REFERENCE DESIGNATIONS MVD-22 20kV Central WTP Area	Scale	NTS	= H01
				Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& 304
00	Issued for Approval	15.01.19	YR	Checked	AW			PT. ABB Sakti Industri		SCHEMATIC DIAGRAM	Drawing Nbr.	18.042-MNA-304.04	Sh. 5
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				SBC-W INCOMING	Customer Doc. Nbr.		Cont. 28

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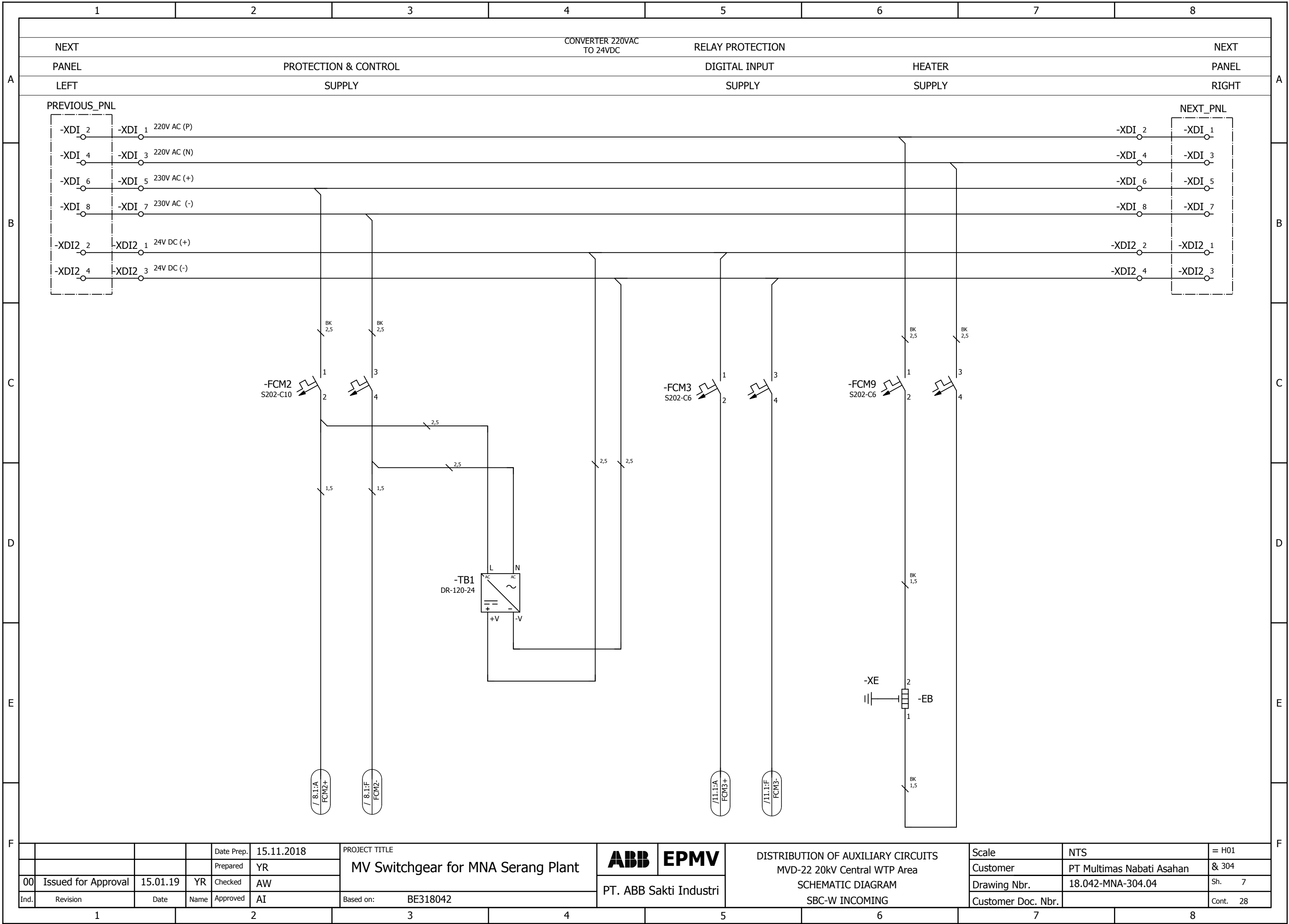
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A	-TFA	ACTIVE POWER TRANSDUCERS	A
	-TFC	CURRENT TRANSDUCERS	
	-TFF	FREQUENCY TRANSDUCERS	
	-TFJ	ACTIVE ENERGY TRANSDUCERS	
	-TFK	REACTIVE ENERGY TRANSDUCERS	
	-TFM	MULTIFUNCTION TRANSDUCERS	
	-TFP	POWER-FACTOR TRANSDUCERS	
	-TFQ	REACTIVE POWER TRANSDUCERS	
	-TFS	SIGNAL CONVERTERS	
	-TFV	VOLTAGE TRANSDUCERS	
B	-WA	BUSBARS	B
	-WBC	POWER CABLES	
	-WE	EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.	
	-WGC	CONTROL CABLES	
	-WGS	COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)	
C	-WH	OPTICAL FIBRES	C
	-XDA	TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS	
	-XDA1	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1	
	-XDA2	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2	
	-XDA3	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3	
	-XDA5	CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER	
	-XDB	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS	
	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)	
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER	
	-XDB10...-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS	
D	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS	D
	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR	
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR	
	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR	
	-XDB95	CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET	
	-XDC	CUSTOMER TERMINAL BLOCK	
	-XDC1	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER	
	-XDC2	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR	
	-XDC3	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT	
	-XDC4	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS	
E	-XDC5	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER	E
	-XDE	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE	
	-XDE1	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE	
	-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2	
	-XDE3	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2	
	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS	
	-XDF1	CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR	
	-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS	
	-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS	
	-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)	
F	-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)	F

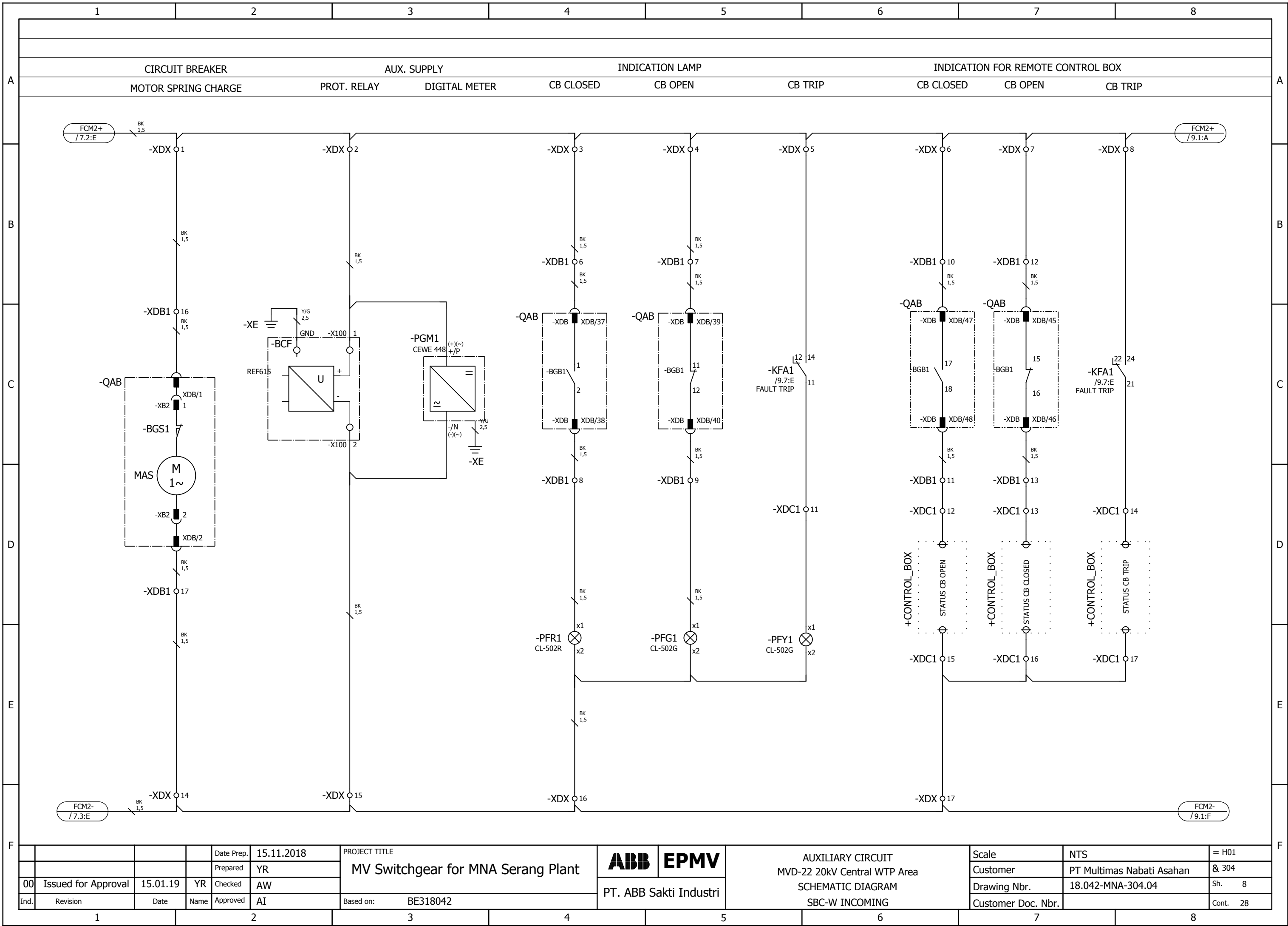
				Date Prep.	15.11.2018	PROJECT TITLE				REFERENCE DESIGNATIONS MVD-22 20kV Central WTP Area	Scale	NTS	= H01
				Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& 304
00	Issued for Approval	15.01.19	YR	Checked	AW						Drawing Nbr.	18.042-MNA-304.04	Sh. 6
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042		PT. ABB Sakti Industri		SBC-W INCOMING	Customer Doc. Nbr.		Cont. 28

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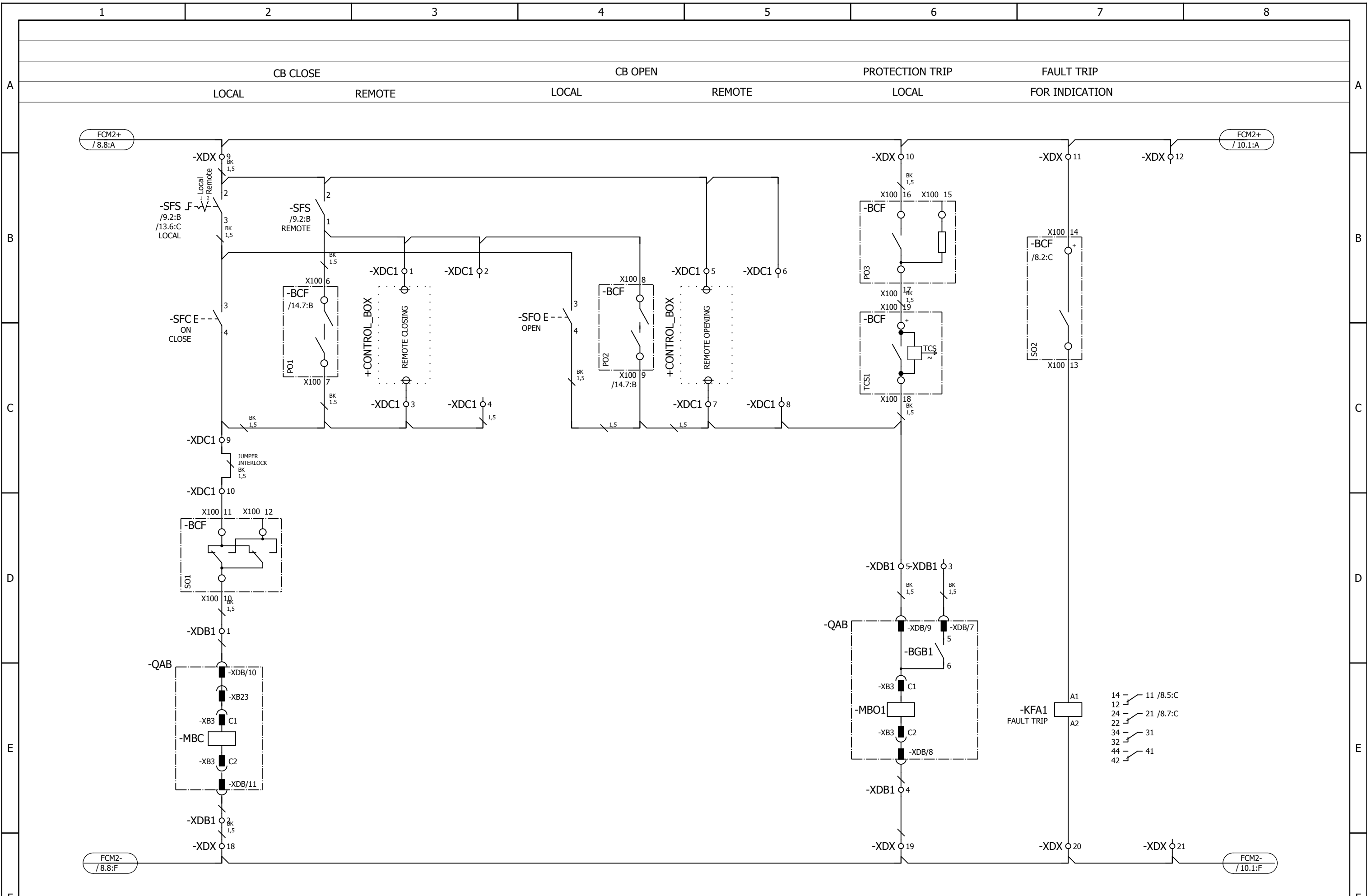
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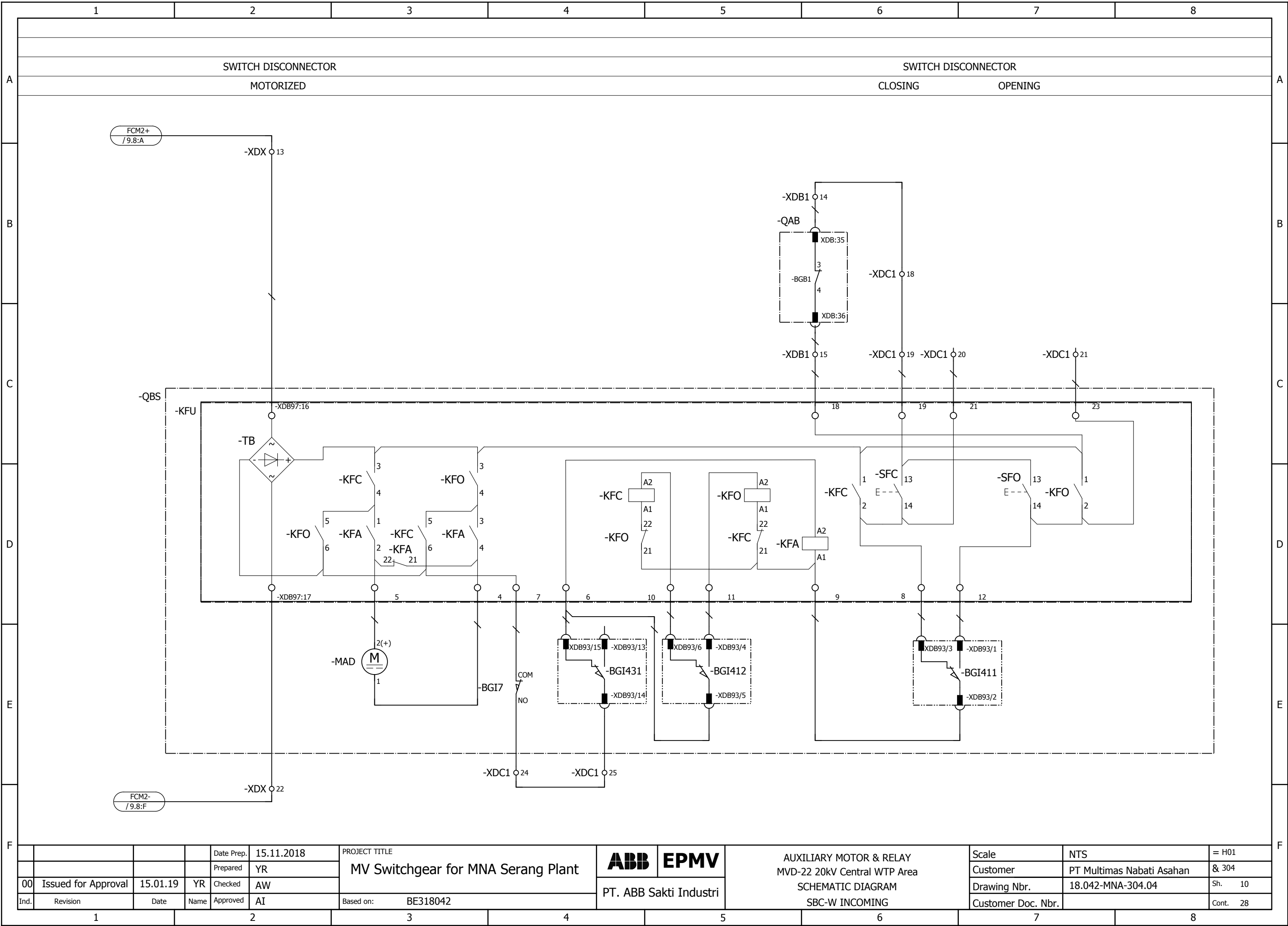


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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	CIRCUIT BREAKER CONTROL MVD-22 20kV Central WTP Area SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 304
00	Issued for Approval	15.01.19	YR	Checked	AW		Drawing Nbr.	18.042-MNA-304.04		Sh. 9		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 28

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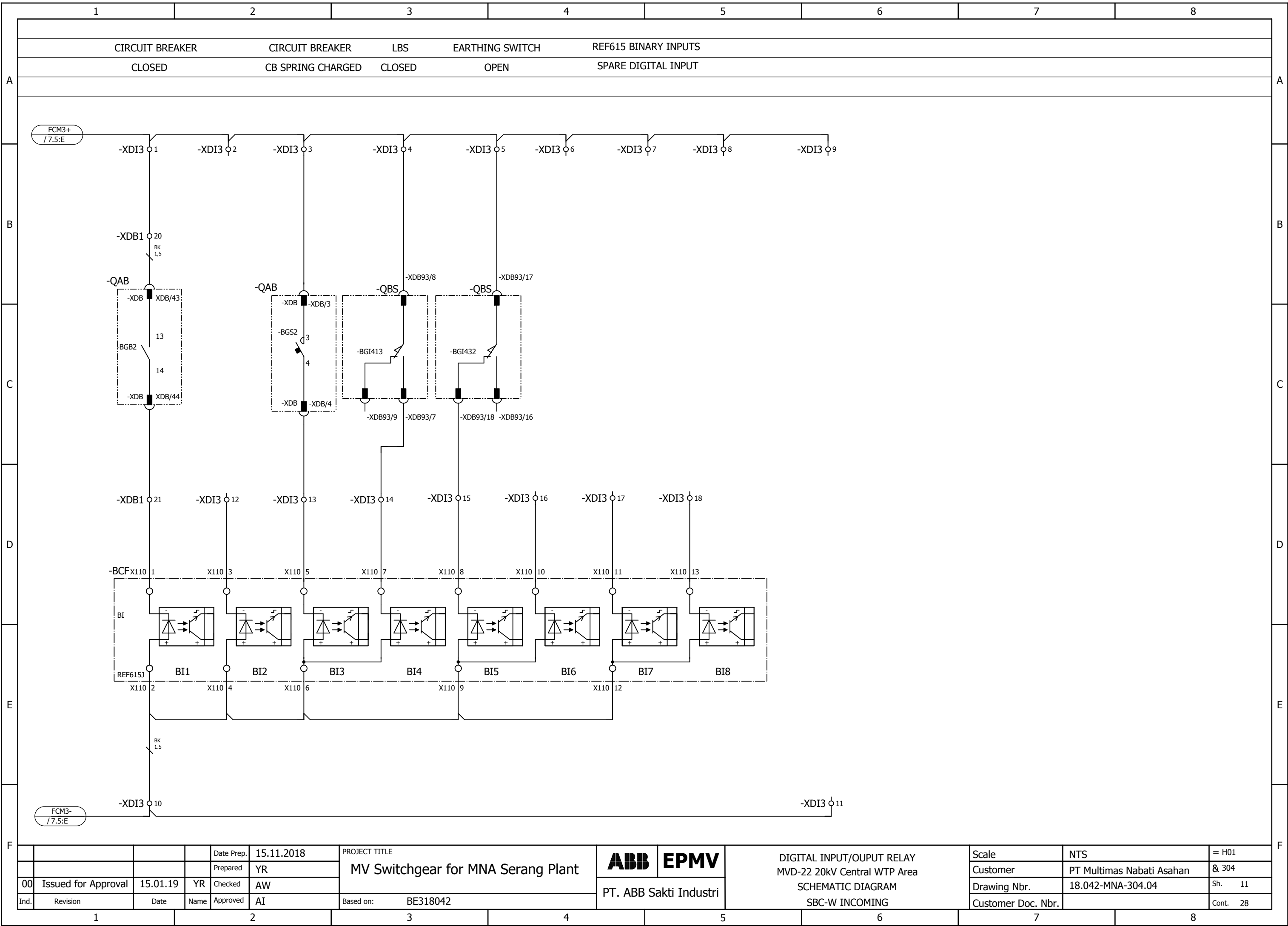


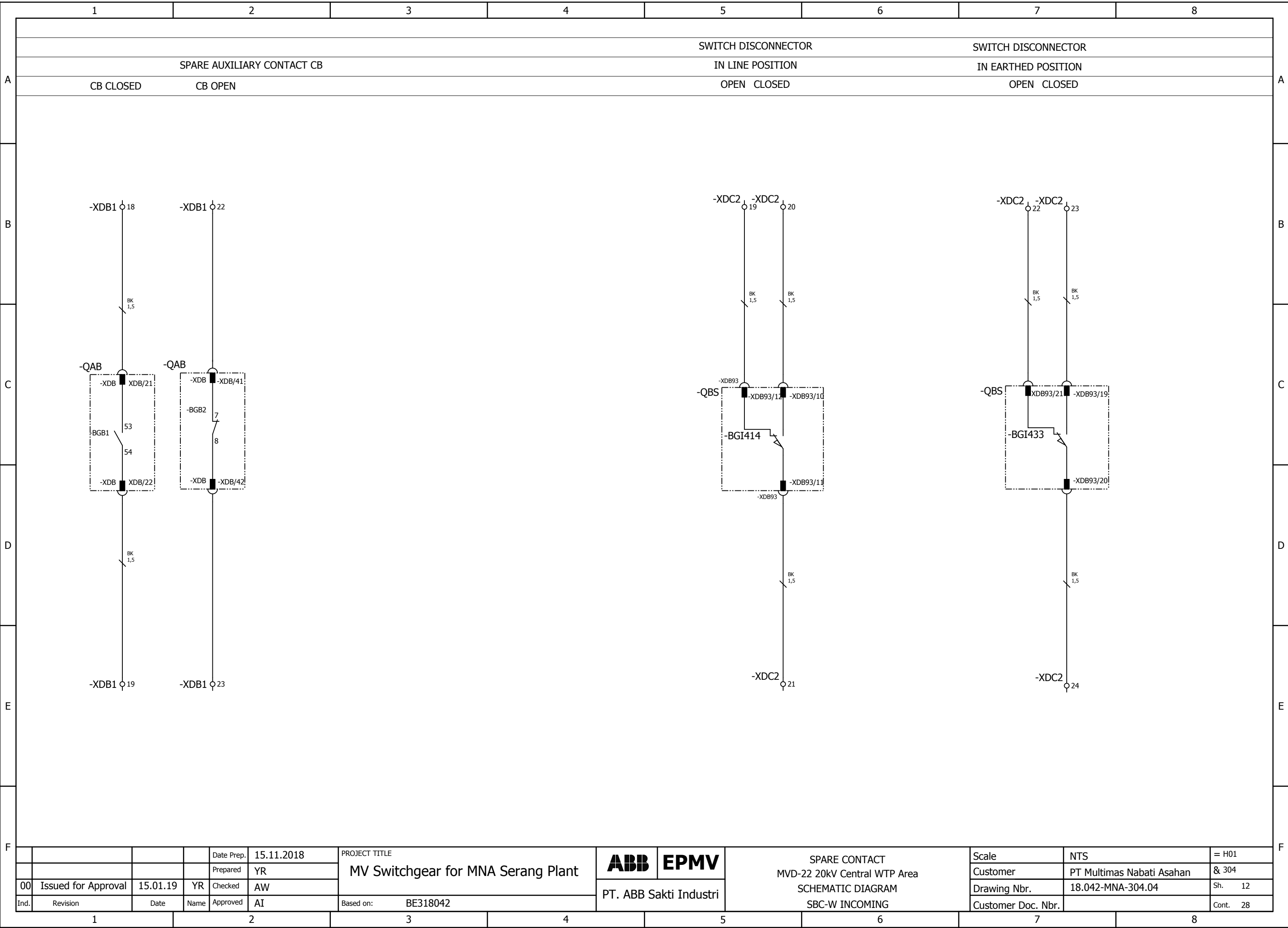
ABB **EPMV**

PT. ABB Sakti Industri

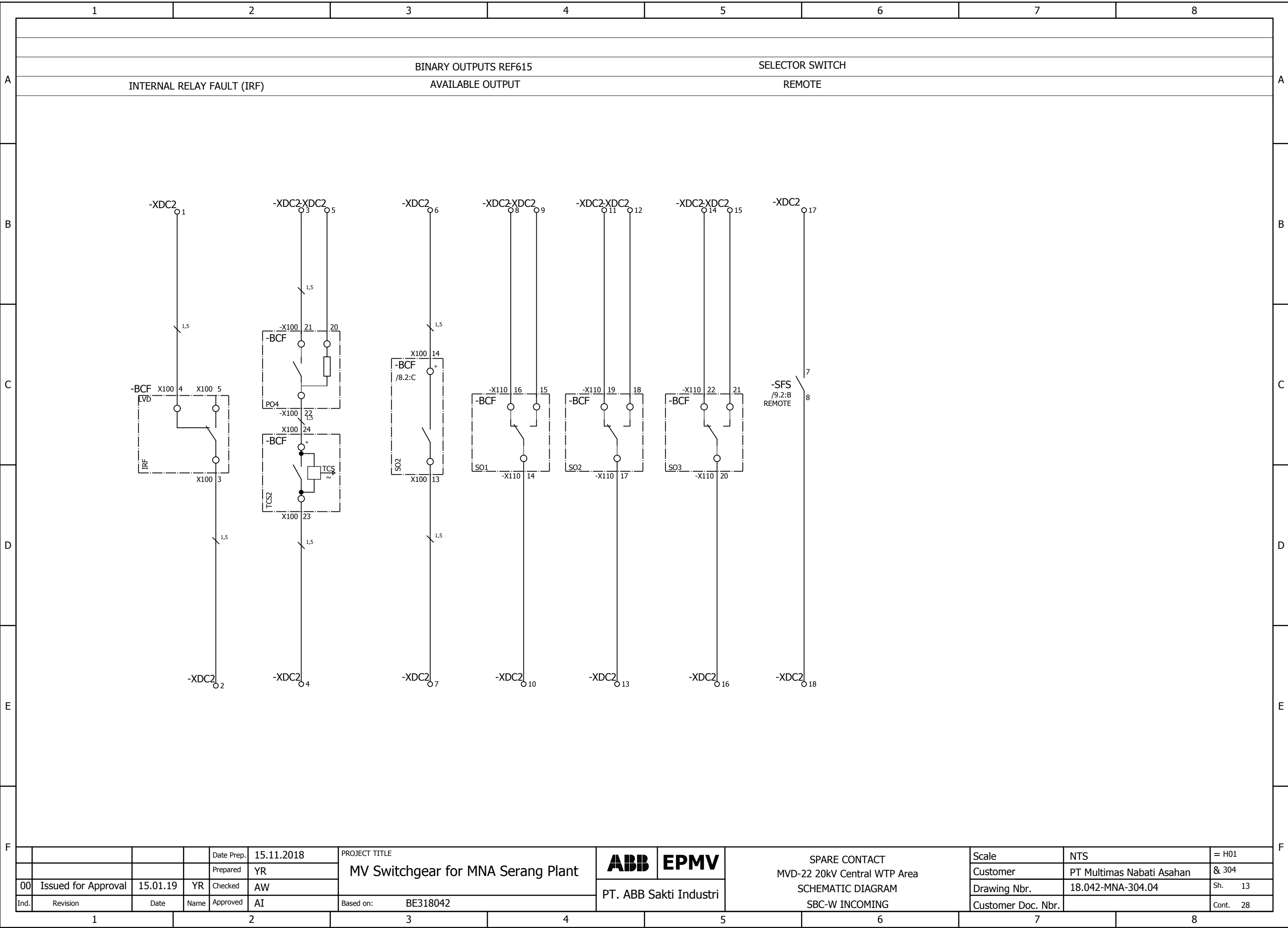
DIGITAL INPUT/OUPUT RELAY
MVD-22 20kV Central WTP Area
SCHEMATIC DIAGRAM
SBC-W INCOMING

Scale	NTS	= H01
Customer	PT Multimas Nabati Asahan	& 304
Drawing Nbr.	18.042-MNA-304.04	Sh. 11
Customer Doc. Nbr.		Cont. 28

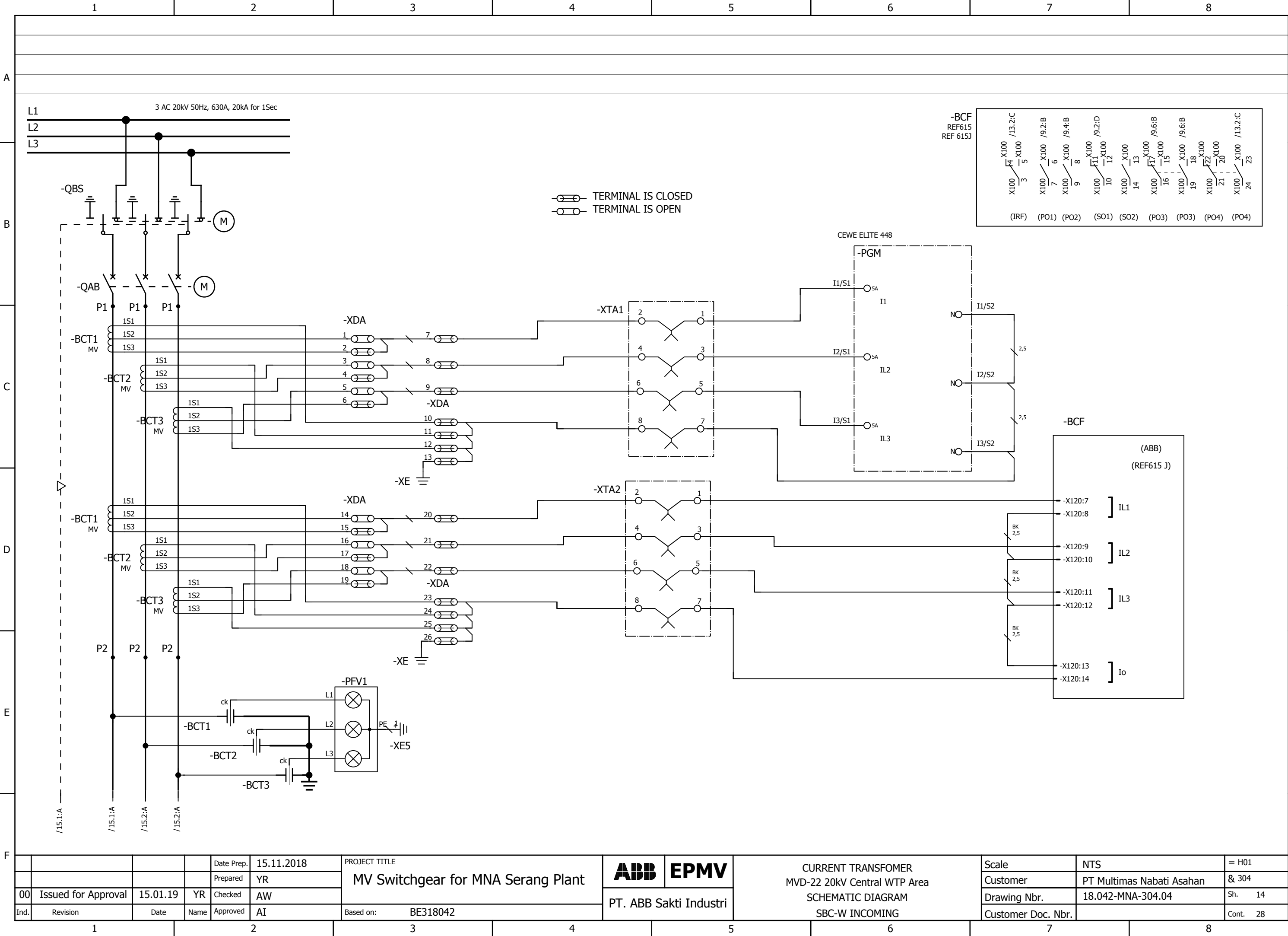
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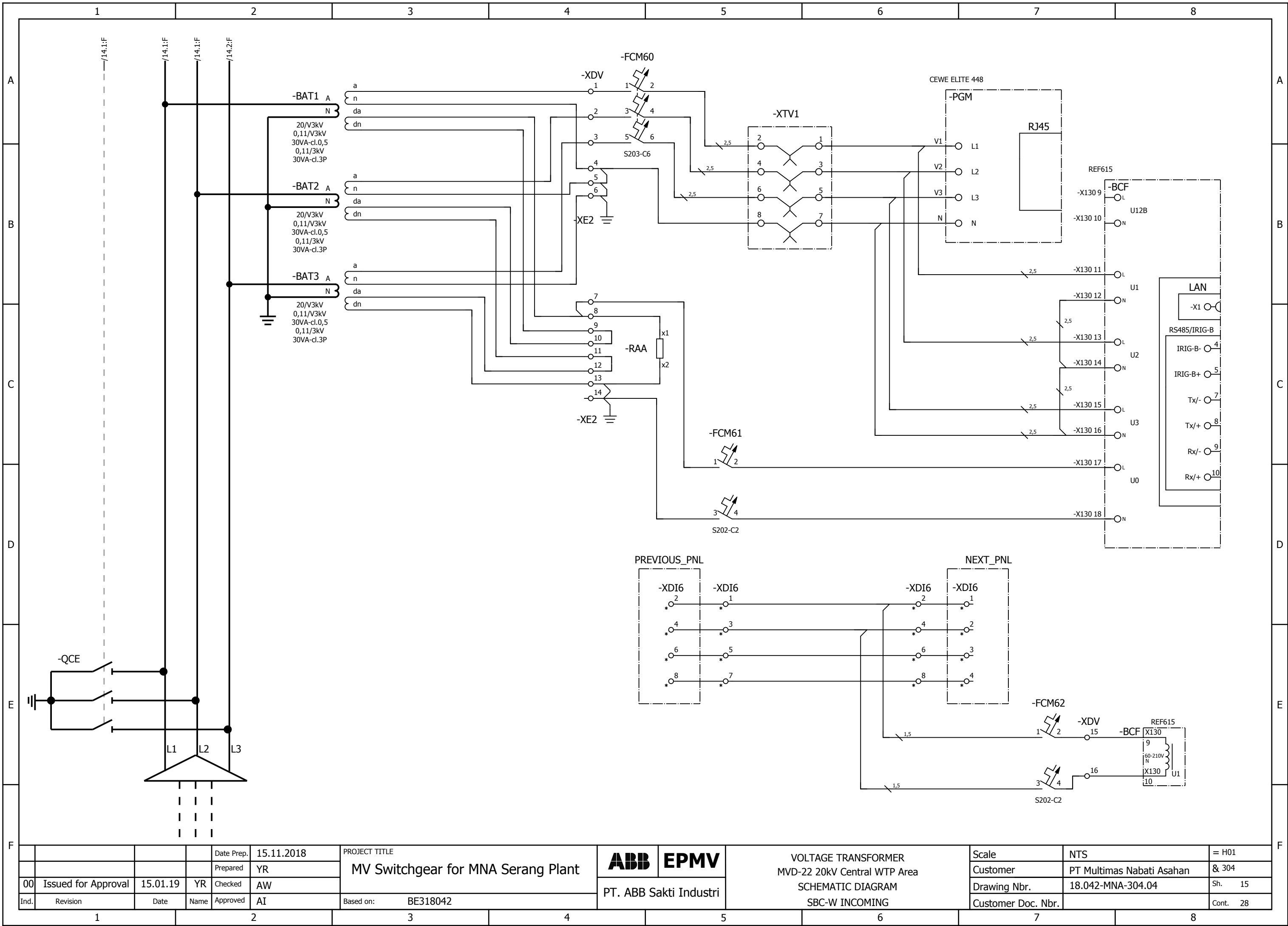
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SPARE CONTACT MVD-22 20kV Central WTP Area SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR							Customer
00	Issued for Approval	15.01.19	YR	Checked	AW					Drawing Nbr.	18.042-MNA-304.04	Sh. 13
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 28

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				Date Prep.	15.11.2018
				Prepared	YR
00	Issued for Approval	15.01.19	YR	Checked	AW
Ind.	Revision	Date	Name	Approved	AI

PROJECT TITLE	
MV Switchgear for MNA Serang Plant	
Based on:	BE318042

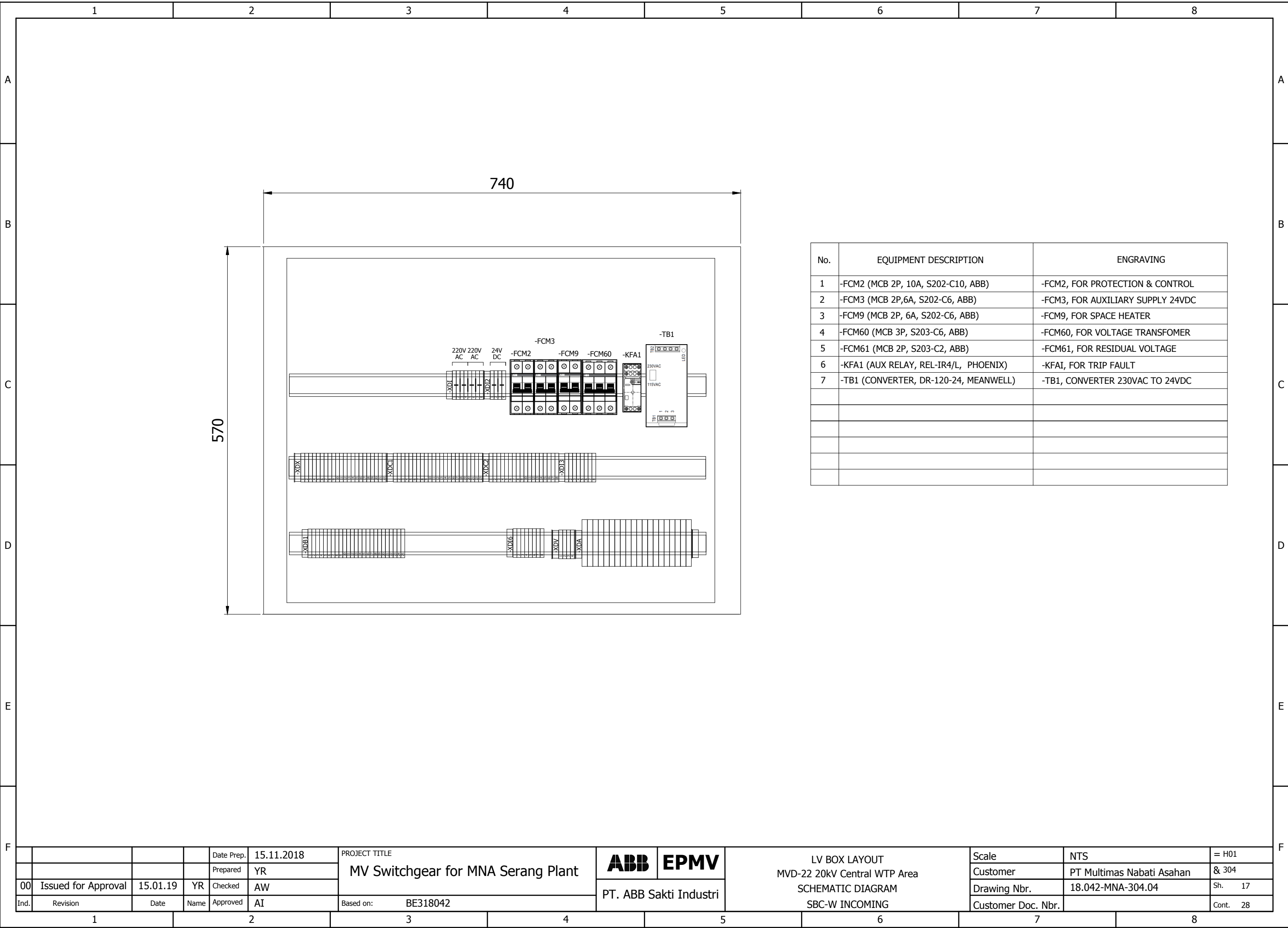
ABB	EPMV
PT. ABB Sakti Industri	

VOLTAGE TRANSFORMER
MVD-22 20kV Central WTP Area
SCHEMATIC DIAGRAM
SBC-W INCOMING

Scale	NTS	= H01
Customer	PT Multimas Nabati Asahan	& 304
Drawing Nbr.	18.042-MNA-304.04	Sh. 15
Customer Doc. Nbr.		Cont. 28

No.	EQUIPMENT DESCRIPTION	ENGRAVING
1	-SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)	-SFS, LOCAL-OFF-REMOTE
2	-SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)	-SFC, CB CLOSE
3	-SFO (PUSH BUTTON CB OPEN, RED, ABB)	-SFO, CB OPEN
4	-PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB)	-PFY1, CB TRIP
5	-PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)	-PFR1, CB CLOSE
6	-PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)	-PFG1, CB OPEN
7	-PGJ (DIGITAL MEASUREMENT, CEWE 448)	-PGJ, DIGITAL MEASUREMENT
8	-BCF (FEEDER PROTECTION RELAY, REF615J, ABB)	-BCF, RELAY PROTECTION
9	-XTA1, XTA2 (TEST BLOCK , YONGSUNG)	-XTA, CURRENT TEST BLOCK
10	-XTV1 (TEST BLOCK , YONGSUNG)	-XTV, VOLTAGE TEST BLOCK

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				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	LV BOX LAYOUT MVD-22 20kV Central WTP Area SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& 304
00	Issued for Approval	15.01.19	YR	Checked	AW			PT. ABB Sakti Industri	Drawing Nbr.		18.042-MNA-304.04	Sh.	17
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042		Customer Doc. Nbr.			Cont.	28

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	1	2	3	4	5	6	7	8
A	Summarized parts list							IDABB_F02_005-IDABB
								Carry over: 0.00
	TAG ID		DESCRIPTION		MANUFACTUR/BRAND	PART NUMBER	QUANTITY	REMARKS
	-BCF		Feeder Management Relay HBFJAEAGABC1ANA11G		ABB	REF615J HBFJAEAGABC1ANA11G	2	/8.2:C;/15.7:B
	-EB		HEATER 230VAC, 50W		SANYANG	CDXN010033P0009	1	/7.6:E
	-FCM2		MCB S202-C10 MCB. 2P, 10A		ABB	2CDS252001R0104	1	/7.2:C
	-FCM3;-FCM9		MCB S202-C6 MCB. 2P, 6A		ABB	2CDS252001R0064	2	/7.5:C;/7.6:C
	-FCM60		MCB S203-C6 MCB. 3P, 6A		ABB	2CDS253001R0064	1	/15.4:A
	-FCM61;-FCM62		MCB S202-C2 MCB. 2P, 2A		ABB	2CDS252001R0024	2	/15.5:D;/15.7:E
	-KFA1		Auxiliary Relay 230AC 4COS		PHOENIX	1YRA3DL266230001	1	/9.7:E
	-PFG1		PILOT LIGHT GREEN #CL-523G#230VAC CL-523G		ABB	1SFA619402R5232	1	/8.5:E
	-PFR1		PILOT LIGHT RED #CL-523R#230VAC CL-523R		ABB	1SFA619402R5231	1	/8.4:E
	-PFY1		PILOT LIGHT YELLOW #CL-523Y#230VAC CL-523Y		ABB	1SFA619402R5233	1	/8.6:E
	-PGM1		POWER METER CEWE 448 Class : 0.5s; CT Input : 1A; VT Input : 57.7-240V (L-N); Aux. Voltage: 80-300VAC/DC		CEWE	1YRA4EA0350001	1	/8.3:C
	-SFC		PUSH BUTTON, YW1B-M1E10-G GREEN COLOUR		IDEC	1YRA3DA14700001	1	/9.2:B
	-SFO		PUSH BUTTON, YW1B-M1E10-R RED COLOUR		IDEC	1YRA3DA14700002	1	/9.4:B
	C	-SFS		LOCAL/REMOTE Selector Switch Type : S0-CA10-A723-600-E; 4NC (for LOCAL Position) & 4NO (for REMOTE Position)		K&N	1YRA3DA17940011	1
-TB1		Voltage Input : 85-264 VAC/120-370V DC Voltage Output : 24 VDC, 5A		MEANWELL	DR-120-24	1	/7.3:D	
-XTA1;-XTA2		Current Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting		YONGSUNG	YSCTT-04C	2	/14.4:B;/14.4:D	
-XTV1		Voltage Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting		YONGSUNG	YSPTT-04C	1	/15.5:A	
D								
E								
F								

				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	PART LIST MVD-22 20kV Central WTP Area SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& 304
00	Issued for Approval	15.01.19	YR	Checked	AW			Drawing Nbr.	18.042-MNA-304.04		Sh. 18		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042		Customer Doc. Nbr.			Cont. 28		

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